

EXPERIENCE

VieON – DatViecVAC | Senior Backend Engineer
 Nov 2025 – Present

- Architected and operated **LiveTV infrastructure** on Flussonic (Ubuntu-based VMs); sustained **~500 MB/s inbound / ~4 Gb/s outbound** traffic, **~60K peak CCU / ~20K daily-avg CCU** across **10+ servers** and **~200 live channels**.
- Cut **~\$1,400/month** (~35M VND) of recurring **Flussonic** license cost by building and adopting **Open Streamer** — a self-built open-source media server.
- Built and operated the **Live Streaming Platform** as core reusable infrastructure — scaled to **700K peak concurrent viewers** powering multiple internal products and editorial workflows.
- Designed and shipped the **Playlist Service** (CDN routing & optimization layer) — routes requests to the nearest, most cost-efficient CDN, reducing cost, latency, and origin load.
- Designed an **in-house transcoding system** processing **6 profiles / ~4 TB raw video per day**, balanced for cost and throughput.
- Developed the **User Service** for unified end-user and internal-user identity, implementing OAuth 2.0 and SSO.
- Set up end-to-end **monitoring & alerting** with anomaly detection plus auto-scaling to absorb dynamic traffic spikes.
- Built **internal admin platforms** for LiveTV, Live Streaming, User Management, and Transcoding operations.

VNPAY | Senior Backend Engineer
 May 2024 – Sep 2025

- Built **scalable microservices** on **gRPC, MySQL, Elasticsearch**, applying **Saga, message queues, 2PC**.
- Built **Monitoring-as-a-Service** with a custom **OpenTelemetry Collector** processing **10B+ records/day**; **Mimir** and **Loki** for metric & log storage; **APISIX** for auth and tenant routing.
- Designed and implemented **VPC Peering** and **Service Endpoints** on **Juniper / OpenStack** — high-speed private connectivity between VPCs and managed services (**DBaaS, S3**).
- Built a **network monitoring system** (**Goflow2, ClickHouse, and Grafana**) ingesting **~5 GB/day** of compressed telemetry for capacity and SLO observability.
- Developed **Database-as-a-Service**: customer-facing feature orchestrating DB clusters on **Kubernetes** with replication and sharding.
- Key contributor to the open-source **terraform-provider-vnpaycloud** GitHub repository.

FPT Telecom | Senior Backend Engineer
 Jun 2022 – May 2024

- Designed and built a **microservices architecture** supporting **~5,000 CCU**.
- Developed a **push notification service** on **Firebase Cloud Messaging**, handling **100K+ messages/day**.
- Built **ETL workflows** on **Airflow** to process **~5 TB/day** of internal data from multiple data warehouses.
- Implemented fine-grained **database security policies** using **OLS** and **VPD** on **Oracle** for access control and data protection.
- Designed and shipped a **user-behavior tracking system** to power analytics and feature-optimization workflows.

EDUCATION

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| B.Sc. Information Technology | University of Science (VNU – HCM) | 3.2/4.0 | 2020 – 2024 |
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TECHNICAL PROFICIENCY

- Programming languages:** Golang, Python, JavaScript, Bash Script.
- Databases:** MySQL, Redis, MongoDB, Elasticsearch, Clickhouse.
- Frameworks/Development packages:**
  - Backend:** Django, Go-gRPC, Gin, ExpressJS.
  - Frontend:** React.
  - DevOps:** Terraform, Grafana, Kubernetes, Consul, ELK Stack, Jenkins, Vault, OpenTelemetry, ...
- BaaS/Cloud services:** GCP, Byteplus.
- Others:** Kafka, Ubuntu, Flussonic.

OTHERS

- Open Streamer** — **High-availability live media server** with multi-protocol ingest/egress and **~150 ms failover**.  
github.com/ntt0601zcoder/open-streamer
- Flussonic Exporter** — Lightweight **Prometheus exporter** for **Flussonic Media Server** with real-time monitoring.  
github.com/ntt0601zcoder/Flussonic-Exporter
- More projects on my portfolio — thuannt.id.vn